higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

12 APRIL 2019

NATED GC

This marking guideline consists of 6 pages.

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SECTION A

QUESTION 1

1.1
1.1.1 obstruction
1.1.2 Velocity
1.1.3 Decrease
1.1.4 Impinges
1.1.5 Pressure energy
1.1.6 Greater
1.1.7 Impact pressure
1.1.8 Static pressure
1.1.9 Tapping
1.1.10 Difference
1.1.11 Impact pressure

(11 × 1) (11)

1.2 The purpose of primary measuring devices is to create a difference in pressure in a flowing fluid. A secondary device then measures this difference in pressure to determine the flow velocity or volume flow rate of the fluid. (4)

1.3
- Flange taps – These are situated 25,4 mm from the upstream and 25,4 mm from the downstream face of the orifice plate.
- Vena contracta taps – The upstream tapping is 1 pipe diameter from the upstream face, and the downstream tapping is determine from a curve relating the required dimension to the ratio of orifice to pipe diameter.
- Pipe taps – These may be 21/2 pipe diameter upstream and 78 diameter downstream from the upstream face of the orifice plate.
- Radius taps (D and D/2 taps) – The upstream pressure tapping is taken at one pipe diameter, D, upstream from the face of the orifice, and downstream pressure tapping is taken at one half pipe diameter, D/2, downstream from the orifice face, approximately the vena contracta position.
- Corner taps – Corner tappings are taken via holes cut obliquely through a flange or boss on the pipe, bringing the inside opening of the holes.(5 × 2) (10)

1.4
- Orifice
- Venturi
- Pitot tube
- Dall tube
- Annubar (5)

[30]

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QUESTION 2

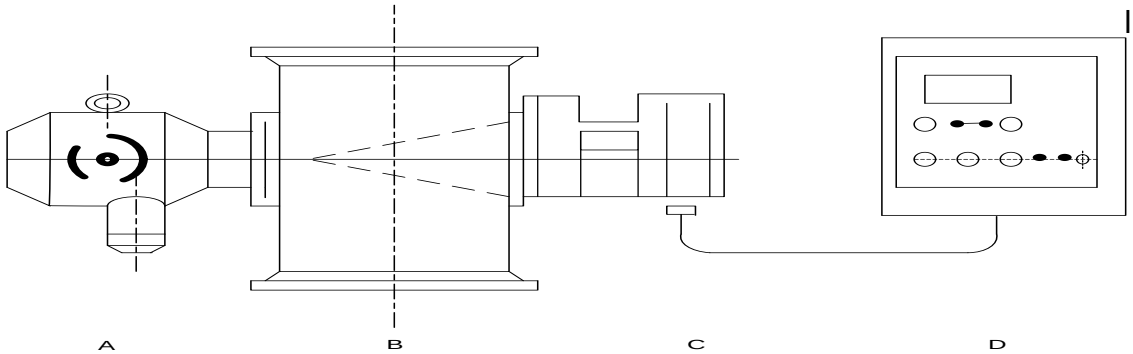
$PE_1 + KE_1 + PV_1 + I_1 = PE_2 + KE_2 + PV_2 + I_2$
$Mgz_1 + mv^2_1/2 + PV_1 + I_1 = mgz_2 + mv_2^2/2 + PV_2 + I_2$
If temperature is constant I = I for liquids only
$V_1 = V_2 = M_1/d_1 = M_2/d_2$ because density = mass x volume
$Mgz_1 + mv_1^2/2 + P_1 m/d = mgz_2 + mv_2^2/2 + P_2 m/d$
Divide the equation by mg
$Z_1 + V_1^2/2g\ P_1/dg = Z_2 + V_2^2/2g + P_2/dg$
But $(Z_1 + P_1/gd) - (Z_2 + P_2/gd) = V_2^2/2g - V_1^2/2g$
$H = V_2^2/2g - V_1^2/2g$
$2gh = V_2^2 - V_1^2$
$2gh = V_2^2 - (A_2^2 V_2^2/A_1^2)$
The ratio A/A is known as the area ratio 'm'
$2gh = V_2^2 - V_2^2 m^2$
$2gh = V_2^2(1-m^2)$
$2ghx(1/1-m^2) = V_2^2$
$V = 2gh.\ 1\sqrt{1-m^2}$
The term $1/\sqrt{1-m}$ is known as the velocity of approach factor and is represented by'E'
$V = E\sqrt{2gh}$
$Q = A_1 V_1 = A_2 V_2$
$V = \frac{Q}{A}$
$Q/A = E\sqrt{2gh}$
$Q = A_2 E\sqrt{2gh}$
But C is known as the discharge coefficient and equals the actual volume flowing divided by the theoretical volume
$Q = C A_2 E\sqrt{2gh}$ **[15]**

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SECTION B: (DENSITY, HUMIDITY AND VISCOSITY MEASUREMENT)

QUESTION 3

3.1



A Measuring section
B The radiation detector
C And finally the measuring converter or amplifier which converts the signal
D Receiver from the detector into a standard process signal (6)

3.2

3.2.1	D
3.2.2	F
3.2.3	E
3.2.4	B
3.2.5	C
3.2.6	A

(6 × 1) (6)

3.3

- Avoid vapour entrainment in the sample liquid due to agitation or boiling
- Avoid a turbulent environment
- Calibrate the instrument regularly to take care of the measurements caused by gradual material build-up or wear of the piston and the tube
- The measuring unit should be cleaned at regular intervals
- Size of any undissolved solids in the liquid to be tested should be small enough not to interfere with measurements
- Sensitivity requirements should be known for bleeding process application
- Avoid use of this instrument for liquid having poor flow characteristics

(Any 5 × 1) (5)

3.4

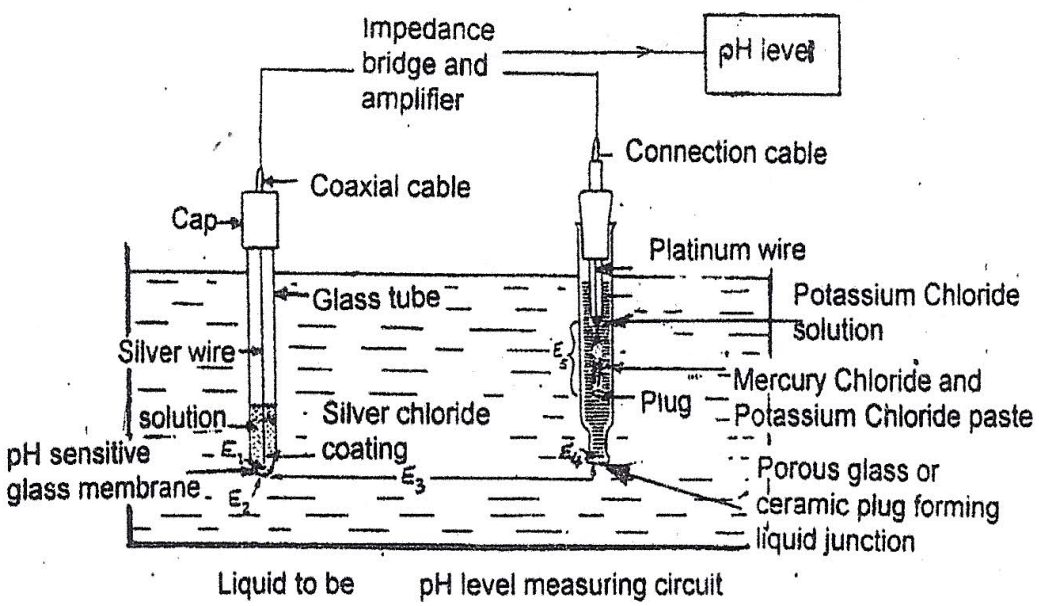
- Dirt on torque measuring devices
- Dirty, corroded, pitted or deformed spindle
- Off-centre placement of spindle in container
- Wrong emf frequency
- Irregular recalibration
- Large air bubble in the sample

(6)
[23]

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SECTION C: (pH MEASUREMENT)

QUESTION 4

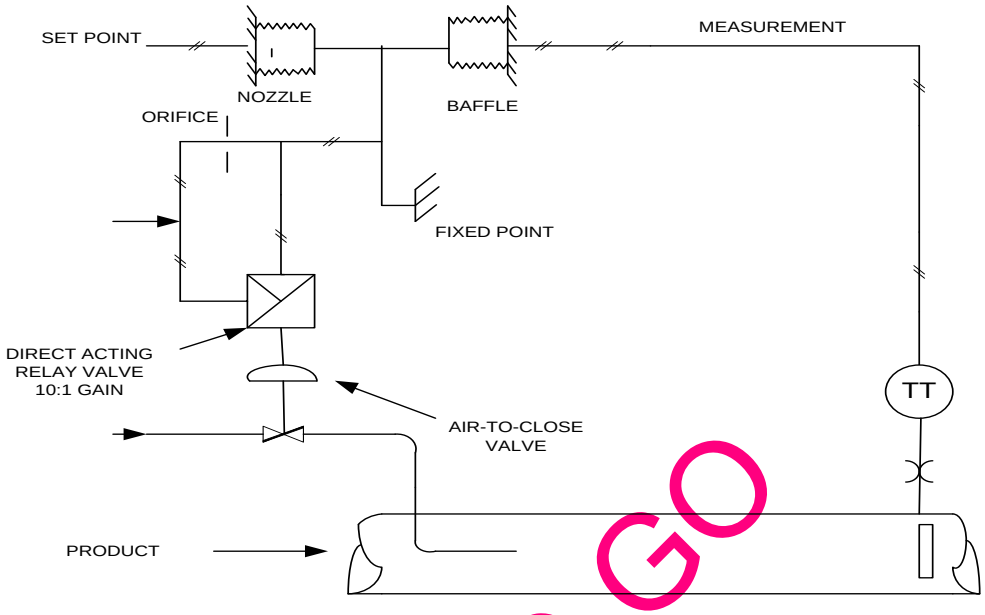


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SECTION D: (AUTOMATIC CONTROL)

QUESTION 5

5.1



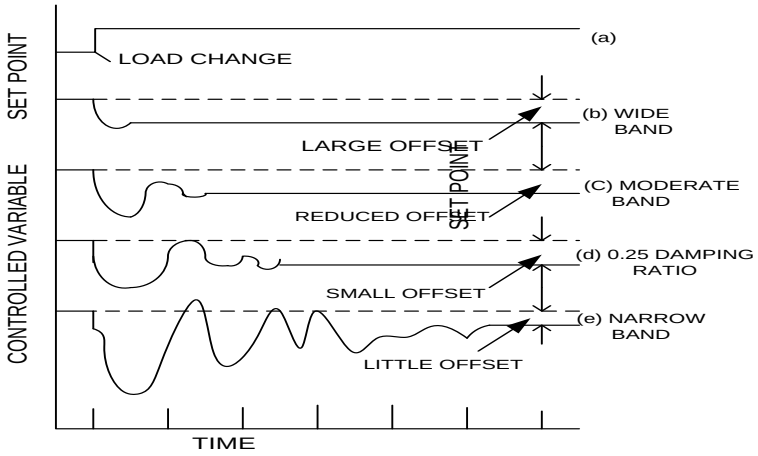
(8)

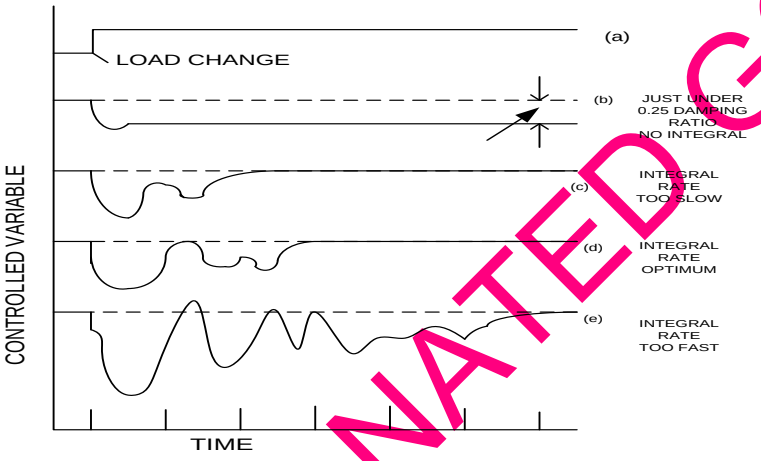
5.2

- Direct acting – An increase in nozzle back pressure resulting an increase in output pressure
- Indirect acting – An increase in nozzle back pressure resulting in a decrease in output pressure
- Non bleed type-An increase in nozzle back pressure resulting in an increase in output pressure (3 × 2) (6)

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5.3





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The first sketch is the correct one (6)

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TOTAL : 100

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